Device-Aware Threshold Personalization for Non-IID Federated IoT Malware Detection

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***Abstract*—Federated IoT anomaly detectors often train one shared autoencoder and then apply one shared reconstruction-error threshold. When physical devices produce different benign traffic patterns, one threshold can concentrate false alarms on a subset of clients. This paper presents a controlled threshold-calibration study—not a new detection algorithm—that isolates threshold scope as the only experimental variable after federated training with $E{=}1$ local epoch and full client participation: the same FedAvg autoencoder, seeds, and score artifacts are reused while shared, per-client, family-based, and cluster-based thresholds are compared.**

On N-BaIoT, where each client is a physical IoT device, per-client calibration (B2) reduces the coefficient of variation of false-positive rate from 1.017 to 0.299 ($\Delta = 0.718$, all five seed-level deltas positive). Cluster-based calibration (B4) recovers approximately 52% of this improvement without requiring a device taxonomy. The gain is not a uniform detection-quality improvement: CV(TPR) increases slightly and the lower-tail Macro-F1 degradation is concentrated in a single low-separability client (Ennio Doorbell), representing a client-specific failure mode rather than a universal tradeoff. On CICIoT2023, where the evaluated file-level pseudo-clients (dataset-file partitions, not physical IoT devices) are near-homogeneous, per-client calibration offers no improvement under this partition; alternative physical-device partitions of CICIoT2023 remain untested. A Dirichlet severity sweep supports the interpretation that threshold personalization is useful mainly under stronger client heterogeneity.

***Index Terms*—federated learning, IoT anomaly detection, threshold calibration, false-positive-rate dispersion, autoencoder, non-IID data, personalization.**

I. Introduction

Federated learning (FL) trains shared models without pooling raw traffic [1]. Reconstruction-error autoencoders (AEs) are a natural fit for IoT intrusion detection because benign-only training suffices and attack labels are not required at training time.

A standard deployment trains one shared AE via FedAvg, then applies a threshold τ above which a reconstruction error signals an intrusion. In homogeneous fleets this works well. In heterogeneous fleets, however, benign reconstruction-error distributions differ substantially across device types: a single shared τ imposes unequal false-positive rates (FPRs). Some devices carry a disproportionate false-alarm burden—making them noisy or operationally unusable—while others may become under-sensitive. A shared server-broadcast threshold is natural in deployments where the system exposes a single global alarm boundary or where per-client threshold state is not managed independently.

We ask how large the per-client FPR disparity is under a shared threshold, how much per-client calibration reduces it, and what detection-quality tradeoff this creates—all under a fixed FedAvg AE with real physical-device heterogeneity. The target operational scenario is a fleet of physical IoT devices for which local benign calibration data are available, and where reducing unequal false-alarm burden across devices is a deployment priority. Prior FL-AE work may report aggregate detection metrics or use thresholds implicitly, but without fixing the encoder, aggregation protocol, and training hyperparameters and varying only threshold scope, it is impossible to attribute per-client alarm-rate dispersion to the threshold policy rather than to training differences. This controlled-comparison gap motivates our study.

The primary metric is the coefficient of variation of FPR across eligible clients, CV(FPR) $= \sigma_{\text{FPR}}/\mu_{\text{FPR}}$. Four threshold policies are evaluated: a client-averaged shared threshold (B1), full per-client personalization (B2), device-family grouping (B3), and unsupervised cluster grouping (B4). B1 is constructed as the arithmetic mean of eligible clients' local 95th-percentile benign calibration thresholds, broadcast as a single scalar. The arithmetic mean treats eligible clients equally and models a simple server-broadcast threshold derived from local calibration summaries, avoiding sample-count domination; pooled and sample-weighted alternatives are reported as sensitivity checks. A centralized baseline (B0) is included as a privacy-incompatible reference comparator but is not part of the single-variable B1–B4 comparison.

The study uses N-BaIoT [2] as the primary confirmatory regime (Regime A), CICIoT2023 [3] as a file-level applicability check under near-homogeneous conditions (Regime B), and synthetic Dirichlet partitions of N-BaIoT as a heterogeneity severity sweep (Regime C). Five random seeds are used; seed-level descriptive summaries (mean, SD, range, sign consistency) report stability. The FL simulation uses Flower [4].

Contributions.

- A controlled threshold-calibration study in which the FedAvg AE ($E{=}1$, full participation), training protocol, and seeds are held fixed across B1–B4; only the post-training threshold scope varies.
- Empirical evidence on N-BaIoT that the client-averaged shared threshold (B1) yields CV(FPR) $\approx$ 1.02; per-client calibration (B2) reduces it to 0.299 ($\Delta = 0.718$, SD $= 0.079$, range $[0.588, 0.772]$, all five seed deltas positive). The detection-quality tradeoff—mean Macro-

TABLE I
POSITIONING: ADAPTATION LAYER AND THRESHOLD TREATMENT.

Work group	Adaptation layer	Threshold treatment
FL-AE IoT IDS [5], [6]	Model (shared AE)	Implicit/global
FedMSE [8]	Aggregation weights	Not varied
pFedMe, FedPer [11], [12]	Model weights/objectives	Not addressed
Kitsune, DIoT [14], [15]	Local AE	Per-device (no controlled study)
DATP (this work)	**Threshold only**	**Controlled comparison**

F1 improves but lower-tail P10 Macro-F1 worsens, concentrated in a low-separability client—is characterized and should be weighed against the FPR-equity gain.
- Applicability conditions: under our CICIoT2023 file-level partition (pseudo-clients, not physical devices), benign distributions are near-homogeneous and per-client calibration offers no improvement. A Dirichlet severity sweep is consistent with the benefit concentrating in high-heterogeneity regimes.

II. Related Work

FL autoencoders for IoT anomaly detection. Rey et al. [5], Khraisat et al. [6], and Olanrewaju et al. [7] apply federated autoencoders to device-partitioned N-BaIoT and report aggregate detection metrics. Thresholds are applied implicitly or globally; none isolates threshold calibration scope as a variable or measures per-client FPR dispersion.

Non-IID aggregation. Nguyen et al. [8] improve AUC under Dirichlet non-IID N-BaIoT by weighting aggregation (FedMSE); the approach modifies the training/aggregation protocol, not the post-training threshold policy, and per-client alarm-rate dispersion is not reported. General non-IID benchmarks [9], [10] focus on loss and accuracy rather than per-client alarm-boundary dispersion.

Model-level FL personalization. Personalization in FL is studied mainly at the model level [11]–[13]: methods such as pFedMe, Per-FedAvg, and FedPer modify model weights or training objectives to adapt to local distributions. These approaches adapt the model parameters; DATP instead fixes the shared AE and adapts only the post-training alarm boundary—the two adaptation layers are complementary.

Threshold-level personalization. Per-device thresholds are a natural deployment option acknowledged in Kitsune [14] and in DIoT [15], but neither provides a controlled comparison fixing all other variables. The existence of local thresholds is not new; the missing element is a controlled study that fixes encoder, training protocol, and seeds across threshold-scope variants and then measures per-client FPR dispersion.

Table I summarizes the gap.

III. Problem Formulation

A. Notation

Let $\mathcal{K} = \{1, \ldots, K\}$ denote the set of FL clients. Each client k holds local benign traffic data $\mathcal{D}_k^{\text{ben}}$ and may have attack data $\mathcal{D}_k^{\text{atk}}$. An autoencoder $h = g \circ f$ maps input $\mathbf{x} \in \mathbb{R}^d$ to a reconstruction $\hat{\mathbf{x}} = g(f(\mathbf{x}))$, where f is the encoder and g the decoder. The reconstruction error is the per-sample mean squared error:

$$e(\mathbf{x}) = \frac{1}{d}\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2, \quad (1)$$

where d is the input dimension.

B. Binary Alarm Decision

Given a threshold τ_k for client k, the alarm decision is:

$$\text{anomaly}(\mathbf{x}) = \mathbf{1}[e(\mathbf{x}) > \tau_k]. \quad (2)$$

The FPR for client k is $\rho_k = \Pr[e(\mathbf{x}) > \tau_k \mid \mathbf{x} \sim \mathcal{D}_k^{\text{ben}}]$.

C. Dispersion Metric

The primary metric is the coefficient of variation of FPR over eligible clients $\mathcal{K}_{\text{elig}} \subseteq \mathcal{K}$:

$$\text{CV(FPR)} = \frac{\sigma_\rho}{\mu_\rho}, \quad \mu_\rho = \frac{1}{|\mathcal{K}_{\text{elig}}|} \sum_{k \in \mathcal{K}_{\text{elig}}} \rho_k, \quad (3)$$

where σ_ρ is the standard deviation. CV(FPR) normalizes cross-client FPR dispersion by the fleet mean, making dispersion comparable across regimes with different mean FPR levels; we report IQR(FPR) and max–min FPR as absolute-dispersion checks to guard against small-denominator artifacts.

Eligibility and fallback. Eligible clients $\mathcal{K}_{\text{elig}}$ are those with at least $n_{\min}$=100 benign calibration samples. Clients below this threshold are *calibration-pending*: they receive the client-averaged shared threshold τ_{global} (B1 value) as a fallback and are excluded from dispersion metrics. Coverage is reported as $|\mathcal{K}_{\text{elig}}|/|\mathcal{K}|$ for all conditions.

Operational assumption. Percentile calibration assumes each client's benign reconstruction-error distribution is sufficiently stable between calibration and test. Temporal drift would weaken the calibration guarantee; this paper does not model concept drift.

D. Confirmatory Statistic

The primary statistical estimate is the seed-level delta:

$$\Delta_s = \text{CV(FPR)}_{\text{B1},s} - \text{CV(FPR)}_{\text{B2},s}, \quad (4)$$

aggregated over seeds $s \in \mathcal{S}$ via descriptive summaries: mean, standard deviation, range, and sign consistency. With only five seeds, we report these statistics as a seed-level consistency summary—not a high-powered population significance test. We interpret all-positive seed deltas as seed-level support for RQ2 in this tested protocol.

E. Research Questions

- **RQ1:** Under the client-averaged shared threshold (B1), how large is the variation of FPRs across heterogeneous IoT clients?
- **RQ2:** Does per-client threshold calibration (B2) reduce CV(FPR) compared with the client-averaged shared threshold, and is the reduction consistent across seeds?
- **RQ3:** How much of the B2 dispersion reduction can cluster-mean calibration (B4) recover?

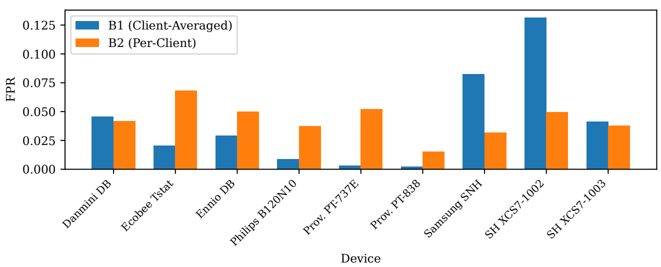

Fig. 1. Per-device FPR under B1 (Client-Averaged Shared τ) and B2 (Per-Client τ) for the nine N-BaIoT clients (seed 0; illustrative preview). Multi-seed inference uses the aggregate results in Table IV.

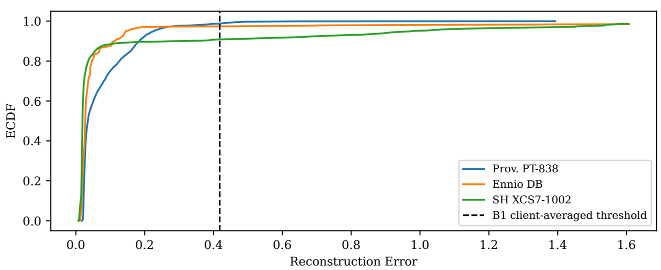

Fig. 2. Benign reconstruction-error empirical cumulative distributions for three representative N-BaIoT clients (seed 0). The shared threshold τ_{global} (B1, dashed) crosses each curve at a different exceedance fraction $(1-F(\tau))$, producing unequal FPRs. Per-client thresholds (B2) set each client's p_{95} independently, aligning benign-tail exceedance rates by construction. Attack distributions are not shown; they need not shift in parallel with benign tails, explaining the TPR/Macro-F1 tradeoff observed in Table IV.

The benign calibration-error distributions of different device types differ substantially in location and spread, so the B1 client-averaged threshold τ_{global} crosses each distribution at a different fraction, producing device-specific FPRs. Per-client percentile calibration is expected to align benign-tail exceedance rates when calibration and test distributions are sufficiently stable; the empirical questions are therefore the magnitude of the FPR-dispersion reduction, the detection-quality tradeoff it creates, and the extent to which grouped calibration (B4) approximates per-client calibration without a device taxonomy—all under a fixed FL-AE setting with real physical-device heterogeneity.

IV. Threshold Personalization Framework

Fig. 1 illustrates the operational consequence of heterogeneous calibration: under B1 (client-averaged shared τ), per-device FPRs vary substantially across the nine N-BaIoT clients, while B2 (per-client τ) reduces this dispersion for the representative seed shown. As an illustrative preview of the primary Regime A pattern, Fig. 1 shows seed 0 only; inference uses the multi-seed results in Table IV.

Pipeline. Each client holds benign traffic split into train, calibration, and test partitions; attack data are reserved exclusively for evaluation and are never used to fit or tune thresholds. A shared AE is trained with FedAvg via the Flower framework [4] (identical training across B1–B4; B0 uses centralized pooling). *The AE is trained once per (dataset, regime, seed, α) condition.* After training, each client scores its calibration and test data under the fixed AE; per-client score arrays are written to disk and reused by all threshold policies without retraining. Even when a run is budget-capped at 150 rounds, the same final AE state is shared across all B1–B4 derivations, preserving the threshold-scope-only comparison.

Threshold policies. Four policies operate on the shared score artifacts:

- **B1 (Client-Averaged):** $\tau_{\text{global}} = \frac{1}{|\mathcal{K}_{\text{elig}}|}\sum_{k\in\mathcal{K}_{\text{elig}}} \tau_k^{(95)}$, applied uniformly to all clients. Pooled and weighted variants are sensitivity checks only (Table VI).
- **B2 (Per-Client):** $\tau_k = \tau_k^{(95)}$ applied locally; zero additional uplink communication in a deployment where each client calibrates its threshold locally.
- **B3 (Family-Mean):** arithmetic mean of local thresholds within each device family (Regime A only; requires a device taxonomy).
- **B4 (Cluster-Mean):** k-means on a 4-scalar fingerprint $[\mu_e, \sigma_e, \text{skew}_e, p_{95}(e)]$ (StandardScaler-normalized); K=3 for Regime A, silhouette-selected for Regimes B/C. The four statistics are compact descriptors of the benign calibration-error distribution: μ_e captures location, σ_e spread, skew_e tail asymmetry, and $p_{95}(e)$ the operating tail used by thresholding. They are distributional summaries for grouping, not privacy mechanisms. In Regime A, K=3 is a practical fixed choice: with only 9 physical devices and coarse device-family structure, silhouette-based K selection is unstable at small n; K=3 avoids overfitting cluster count to sampling noise. B4 requires sharing these distributional fingerprints, which carries nonzero metadata leakage risk (Section VII).

Clients with fewer than $n_{\min}$=100 benign calibration samples are calibration-pending; they receive τ_{global} as fallback and are excluded from CV(FPR). **Evaluation.** Each client evaluates FPR, TPR, balanced accuracy, and Macro-F1 on held-out test sets. CV(FPR), CV(TPR), IQR(FPR), worst-client balanced accuracy, P_{10} Macro-F1 (10th percentile of per-client Macro-F1 across eligible clients), and coverage ratio are aggregated. Seed-level deltas are summarized descriptively (mean, SD, range, sign consistency).

V. Experimental Design

A. Datasets and Regimes

Regime A — N-BaIoT (confirmatory). The N-BaIoT dataset [2] contains traffic from nine physical IoT devices grouped by the configured device-family taxonomy, infected with Mirai and BASHLITE botnets. Each physical device is one FL client ($K = 9$). The N-BaIoT natural partition mixes device-specific benign feature heterogeneity with some attack-variant skew, and is therefore treated as a realistic natural partition rather than a pure feature-skew isolation regime. Regime A is the primary planned condition for RQ2.

Regime B — CICIoT2023 (file-level applicability check). CICIoT2023 [3] provides 63 file-defined pseudo-clients under our partition. Each of the 63 file-defined clients corresponds to one merged flow-level traffic file from the dataset's `MERGED_CSV` directory; client boundaries are defined by the dataset's own file-level granularity, not by physical device identity. Within each client, benign records are randomly shuffled (no temporal ordering is preserved) and split 70/15/15 into train, calibration, and test-benign partitions; attack records are reserved exclusively for test-attack evaluation. A per-client cap of 50,000 total records is applied, reserving up to 10,000 attack records. Regime B tests whether the Regime A personalization benefit holds when benign reconstruction-error distributions are near-homogeneous under this partition; it is not part of the primary B1–B4 comparison.

Regime C — Dirichlet non-IID severity (controlled sweep). N-BaIoT traffic is repartitioned using a Dirichlet distribution with $\alpha \in \{0.1, 0.3, 0.5, 1.0, 10.0, \text{IID}\}$ across $K = 20$ synthetic clients. Lower α means higher heterogeneity. Regime C provides secondary controlled non-IID severity evidence for the conditions under which threshold personalization is beneficial.

B. Architecture and Training

The AE encoder has dimensions $[d, 80, 40, 20]$ (ReLU activations, no batch normalization), where $d = 115$ for N-BaIoT and $d = 39$ for CICIoT2023. FL simulation uses the Flower framework [4] with FedAvg for up to 150 rounds and a relative-change convergence criterion (window = 10 rounds, tolerance = 0.005); clients perform E=1 local epoch per round with Adam (lr = 10^{-3}, batch size 256). The choice E=1 keeps training dynamics conservative and fixed while isolating post-training threshold scope; evaluating larger E is future work because it changes client drift and reconstruction-error geometry. With E=1 and full client participation, FedAvg approaches synchronous distributed training with frequent aggregation; we retain the FedAvg framing for consistency with FL-AE literature, and the post-training threshold comparison is invariant to this distinction. B0 trains a centralized AE on pooled benign data, applying the pooled 95th-percentile threshold; it is excluded from the B1–B4 controlled ladder.

C. Evaluation Protocol

Five seeds (0–4) are used. The AE is trained once per seed; B1–B4 thresholds are derived from shared per-client score artifacts. The calibration quantile is $q = 0.95$; clients with fewer than $n_{\min} = 100$ benign calibration samples are calibration-pending, receive τ_{global} as fallback, and are excluded from CV(FPR). Budget-capped runs (150 rounds without satisfying the convergence criterion) reuse the final AE across all B1–B4 derivations, preserving the threshold-only comparison invariant. In Regime A all 5 seeds converged (rounds 51–93); in Regime B all 5 were budget-capped; in Regime C, 21 of 30 seeds converged (9 budget-capped, concentrated at $\alpha \in \{10.0, \text{IID}\}$).

TABLE II
EXPERIMENTAL REGIME SUMMARY

Regime	Dataset	K	Purpose
A	N-BaIoT	9	Confirmatory (B1 vs B2)
B	CICIoT2023	63	Condition-of-applicability check
C	N-BaIoT (Dirichlet)	20	Non-IID severity sweep (secondary)

TABLE III
REPRODUCIBILITY SPECIFICATION.

Parameter	Value
AE encoder dims	$[d, 80, 40, 20]$
Loss / Optimizer	MSE / Adam (lr 10^{-3})
Batch size	256
Local epochs per round	1
Rounds budget	150 (convergence early-stop)
Convergence criterion	rel. change < 0.005, window 10
Client participation	Full (all clients each round)
Normalization scope	Per-client (StandardScaler)
Seeds	$\{0, 1, 2, 3, 4\}$
Threshold quantile q	0.95
Cal. eligibility $n_{\min}$	100 benign cal. samples
N-BaIoT split ratios	60/20/18 (train/cal/test, gapped)[a]
CICIoT2023 split	70/15/15 benign; attack test-only[b]
CICIoT2023 cap	50,000 total per client (10,000 attack)
Score artifact reuse	Train once; B1–B4 from shared scores

[a] Chronological split with two 1%-wide temporal gaps to prevent leakage. Nominal test fraction = 0.18; gap buffers excluded from all partitions.
[b] CICIoT2023: benign split 70/15/15; attack reserved for test only (`calibration_benign_only=True`).

D. Statistical Analysis

The primary statistic is the per-seed delta $\Delta_s = \text{CV(FPR)}_{\text{B1},s} - \text{CV(FPR)}_{\text{B2},s}$. With only five seeds, we report seed-level descriptive summaries—mean, standard deviation, range, and sign consistency—rather than confidence intervals. Coverage ratio $\rho_{\text{cov}} = |\mathcal{K}_{\text{elig}}|/|\mathcal{K}|$ is reported for all conditions.

Table II summarizes the three regimes. Table III provides the full reproducibility specification.

Environment. All experiments ran on a single workstation: AMD Ryzen 5 7600 (6-core), 12 GB RAM, NVIDIA GeForce RTX 5060 Ti (16 GB), Ubuntu 24.04 LTS. Software: Python 3.12.3, PyTorch 2.11.0 (CUDA 13.0), Flower 1.29.0, Ray 2.55.0, NumPy 2.4.4, pandas 2.3.3, scikit-learn 1.8.0. Runtime is not used as an evaluation endpoint in this threshold-scope study.

VI. RESULTS AND DISCUSSION

A. RQ1: FPR Dispersion Under the Client-Averaged Shared Threshold

Table IV presents aggregate results for Regime A over 5 seeds. Under B1, CV(FPR) is 1.017 ± 0.019—per-client FPRs

TABLE IV

REGIME A (N-BAIOT, $K = 9$, COVERAGE 9/9, PENDING 0): MEAN $\pm$ STD OVER 5 SEEDS. B0 IS A CENTRALIZED REFERENCE, EXCLUDED FROM THE B1–B4 CONTROLLED LADDER. WORST BA = WORST-CLIENT BALANCED ACCURACY; MEAN F1 = MEAN MACRO-F1; P10 F1 = 10TH-PERCENTILE MACRO-F1 ACROSS ELIGIBLE CLIENTS.

Baseline	mean FPR	mean TPR	CV(FPR)	CV(TPR)	Worst BA	mean F1	P10 F1
B0 (Centralized)	0.057 ± 0.001	0.829 ± 0.037	0.787 ± 0.031	0.278 ± 0.036	0.653 ± 0.010	0.720 ± 0.054	0.403 ± 0.059
B1 (Client-Averaged)	0.039 ± 0.002	0.738 ± 0.050	1.017 ± 0.019	0.289 ± 0.029	0.649 ± 0.011	0.563 ± 0.068	0.344 ± 0.062
B2 (Per-Client)	0.043 ± 0.001	0.744 ± 0.019	0.299 ± 0.073	0.336 ± 0.007	0.651 ± 0.001	0.604 ± 0.042	0.300 ± 0.000
B3 (Family-Mean)	0.051 ± 0.006	0.745 ± 0.018	0.964 ± 0.096	0.336 ± 0.007	0.653 ± 0.004	0.606 ± 0.038	0.299 ± 0.000
B4 (Cluster-Mean)	0.044 ± 0.003	0.718 ± 0.019	0.645 ± 0.045	0.322 ± 0.012	0.654 ± 0.007	0.545 ± 0.036	0.299 ± 0.000

Primary result: B1$-$B2 CV(FPR) $\Delta = +0.718$, SD$\,=0.079$, seed range $[0.588,\ 0.772]$, all 5 deltas positive; B1$-$B4 $\Delta = +0.372$, SD$\,=0.055$, seed range $[0.319,\ 0.443]$.

vary by more than their mean (a CV above 1 indicates high relative dispersion). The natural N-BaIoT B1 CV(FPR) (1.017) substantially exceeds the IID Dirichlet baseline (0.074), a difference of 0.943, consistent with device heterogeneity contributing to the observed dispersion. B0's CV(FPR) (0.787), although lower than B1's, remains substantially above B2 (0.299), consistent with any single pooled or shared scalar threshold crossing heterogeneous device-error distributions at different exceedance fractions; B0 is included only as a centralized privacy-incompatible reference, not as part of the controlled B1–B4 ladder.

Fig. 3 shows per-client FPR distributions. The worst client (SimpleHome XCS7 security camera) reaches a mean FPR of approximately 0.122 while other clients operate near zero, illustrating the unequal alarm-rate burden imposed by a single shared threshold across heterogeneous devices.

B. RQ2: Does Per-Client Calibration Reduce CV(FPR)?

The primary endpoint is the B1 vs. B2 CV(FPR) delta per seed. Across the five planned seeds, the B1–B2 CV(FPR) deltas are all positive, with mean $\Delta = 0.718$, SD$\,=0.079$, and range $[0.588, 0.772]$ (Table V). We report these seed summaries descriptively rather than as confidence intervals.

Per-client calibration (B2) reduces CV(FPR) from 1.017 to 0.299 (coverage: 9/9, pending: 0), a reduction of 0.718 CV units (70.6% relative), achieved with zero additional uplink since each client computes and applies its own threshold locally. In our experimental implementation, score artifacts and metrics are written centrally for reproducibility analysis; in deployment, B2 requires no threshold upload. B4, by contrast, requires transmitting the 4-scalar fingerprint per client.

The tradeoff with detection uniformity is clear from Table IV. CV(TPR) rises under B2 (0.336 vs. B1's 0.289): narrowing FPR variance modestly widens TPR variance. Worst-client balanced accuracy is comparable (B1: 0.649, B2: 0.651). B2 slightly increases mean FPR relative to B1 (0.043 vs. 0.039), but it redistributes false alarms more evenly; B1 attains a lower fleet-average FPR partly by concentrating false alarms on high-variance clients. Mean Macro-F1 improves (0.604 vs. 0.563), but the lower-tail P_{10} Macro-F1—the 10th percentile of per-client Macro-F1 across eligible clients—falls from 0.344 (B1) to 0.300 (B2). The near-identical P10 Macro-F1 values across B2–B4 ($\approx$ 0.300, std $<$ 0.001) reflect a

TABLE V

PER-SEED CV(FPR) AND DELTAS, REGIME A. ALL $\Delta_{\text{B1-B2}}$ VALUES ARE POSITIVE. B4 RECOVERY = (B1 $-$ B4)/(B1 $-$ B2).

Seed	B1	B2	B4	$\Delta_{\text{B1-B2}}$	B4 recov. (%)
0	1.045	0.347	0.632	+0.698	59.2
1	1.017	0.251	0.693	+0.765	42.3
2	1.015	0.243	0.654	+0.772	46.6
3	1.018	0.251	0.575	+0.767	57.8
4	0.992	0.404	0.673	+0.588	54.3
Mean	1.017	0.299	0.645	+0.718	51.8

floor effect: inspection of per-seed per-client Macro-F1 shows that the Ennio Doorbell client determines the P10 floor under B2–B4 in all seeds, yielding negligible cross-seed variance at this percentile. This is a different failure mode from the SimpleHome XCS7 high-FPR case: XCS7 illustrates shared-threshold false-alarm concentration under B1, whereas Ennio Doorbell drives the lower-tail Macro-F1 after personalization. A post-hoc score-separation audit using existing Regime A score artifacts (averaged over 5 seeds) shows that Ennio Doorbell has the lowest benign-vs-attack AUROC among all nine clients (0.931) and the highest fraction of attack mass below the personalized p_{95} threshold (64.9%), confirming that its attack reconstruction errors substantially overlap with its benign tail. Local thresholding therefore controls Ennio's benign false alarms at the cost of missing the majority of its low-magnitude attack traffic, explaining the P10 Macro-F1 floor.

Mechanism. Per-client 95th-percentile thresholding aligns each client's benign tail, which mechanically equalizes false-alarm rates on benign test traffic. However, attack reconstruction-error distributions need not shift in parallel with benign distributions: when a client has a noisy benign tail, raising its threshold reduces benign false alarms but can also swallow low-magnitude anomaly scores, reducing TPR or lower-tail Macro-F1 (Fig. 2 illustrates the benign-side mechanism). B2 should therefore be interpreted as an FPR-dispersion calibration policy, not a uniform detection-quality improvement.

All five seed-level deltas are positive, ranging from 0.588 to 0.772. Table V provides the full per-seed evidence; we report these descriptively rather than as population-level inference from five seeds.

Absolute FPR dispersion shows the same B1$>$B2 ordering: IQR(FPR) decreases from 0.035 ± 0.002 (B1) to 0.015 ± 0.003 (B2), and the max–min FPR gap narrows from 0.120 ± 0.007 to 0.041 ± 0.012. Because CV(FPR) can be sensitive when mean FPR is small (mean FPR under B1: 0.039; under B2: 0.043), absolute dispersion measures are reported alongside it throughout. For B1, CV(FPR)$\,=1.017$ and mean FPR$\,=0.039$ imply $\sigma_{\text{FPR}} \approx 0.040$; the CV above 1 therefore reflects genuine cross-client spread comparable to the mean, not only a small-

TABLE VI

SHARED-THRESHOLD CONSTRUCTION SENSITIVITY, REGIME A ($n_{\text{SEEDS}} = 5$, $q = 0.95$). NO AE RETRAINING; SENSITIVITY VARIANTS LABELED S. ALL THREE B1 VARIANTS SHOW LARGER CV(FPR) THAN B2, CONSISTENT WITH THE B2 ADVANTAGE NOT BEING DRIVEN SOLELY BY ARITHMETIC-MEAN CONSTRUCTION.

Rule	Construction	CV(FPR)	IQR(FPR)	Max–min FPR	Δ vs B2
B1 (main)	Arith. mean of local p_{95}	1.017 ± 0.019	0.035 ± 0.002	0.120 ± 0.007	$+0.718$
B1-pooled (S)	Pooled p_{95}	0.805 ± 0.048	0.084 ± 0.007	0.134 ± 0.012	$+0.506$
B1-weighted (S)	Sample-wt. mean of local p_{95}	0.907 ± 0.033	0.050 ± 0.018	0.129 ± 0.003	$+0.608$
B2	Per-client p_{95}	0.299 ± 0.073	0.015 ± 0.003	0.041 ± 0.012	—

Mean FPR for the main B1/B2 comparison is reported in Table IV; Table VI focuses on dispersion sensitivity across shared-threshold constructions.

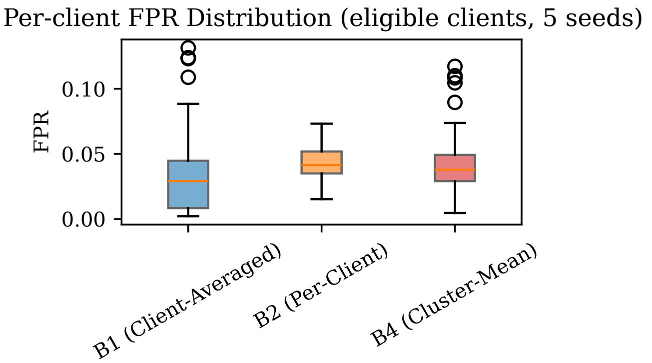

Fig. 3. Per-client FPR distributions under B1, B2, and B4 (Regime A, pooled over 5 seeds). Descriptive only; inference uses seed-level deltas reported in text.

denominator artifact.

Shared-threshold construction sensitivity. Pooled and sample-weighted shared-threshold variants preserve the B2 advantage, suggesting the B1-to-B2 reduction does not appear to be driven solely by the arithmetic-mean construction (Table VI).

C. RQ3: Partial Recovery by Cluster-Mean Calibration

Cluster-mean calibration (B4) achieves CV(FPR) $= 0.645 \pm 0.045$, representing a partial but consistent improvement over B1. All five seed-level B1$-$B4 deltas are positive (mean $\Delta = 0.372$, SD $= 0.055$, range $[0.319, 0.443]$); all five B4$-$B2 deltas are also positive (mean $\Delta = 0.346$, range $[0.269, 0.442]$). B4 is consistently better than B1 but worse than B2, recovering approximately 52% of B2's total improvement (mean-level: $0.372/0.718$; per-seed range: 42%–59%, Table V). B4 requires no device taxonomy and transmits only compact 4-scalar fingerprints per client; however, it does not provide privacy (see Section IV). Its value lies in deployments where per-client state is unavailable and operators need a taxonomy-free intermediate policy—at the cost of distributional metadata leakage. B4's fingerprint still depends on calibration-error statistics including p_{95}, so clients below the eligibility threshold remain calibration-pending and use τ_{global}.

Family-mean calibration (B3, Regime A only) reduces CV(FPR) negligibly (0.964 vs. B1's 1.017): the coarse N-BaIoT device-family labels (doorbell, camera, thermostat) do not align with the reconstruction-error calibration structure—devices within the same family can have dissimilar benign-error distributions, while devices across families can be similar. This taxonomy-independence of calibration structure motivates B4's data-driven clustering over B3's label-based grouping.

TABLE VII

REGIME B (CICIOT2023 FILE-DEFINED CLIENTS, $K = 63$, COVERAGE 63/63, PENDING 0): MEAN $\pm$ STD OVER 5 SEEDS, ELIGIBLE CLIENTS ONLY. B3 NOT APPLICABLE. ALL RUNS BUDGET-CAPPED AT 150 ROUNDS.

Baseline	mean FPR	mean TPR	CV(FPR)	CV(TPR)	Worst BA	mean Macro-F1
B0 (Centralized)	0.051 ± 0.001	0.981 ± 0.000	0.109 ± 0.020	0.001 ± 0.000	0.954 ± 0.006	0.959 ± 0.000
B1 (Client-Averaged)	0.051 ± 0.001	0.982 ± 0.000	0.099 ± 0.002	0.001 ± 0.000	0.959 ± 0.002	0.959 ± 0.000
B2 (Per-Client)	0.051 ± 0.001	0.982 ± 0.000	0.133 ± 0.007	0.001 ± 0.000	0.956 ± 0.001	0.959 ± 0.000
B4 (Cluster-Mean)	0.051 ± 0.001	0.982 ± 0.000	0.101 ± 0.003	0.001 ± 0.000	0.959 ± 0.002	0.959 ± 0.000

D. Regime B: File-Level Applicability Check (CICIoT2023)

Table VII shows Regime B results. Under our file-level CICIoT2023 partition, benign reconstruction-error distributions are near-homogeneous (pairwise JS divergence: mean 0.004, median 0.003, max 0.038 over all $\binom{63}{2}$ pairs and 5 seeds). B1 achieves CV(FPR) $= 0.099 \pm 0.002$; B2 increases it to 0.133 ± 0.007 (all five seed-level B1$-$B2 deltas are negative; mean -0.034, SD 0.006, range $[-0.043, -0.026]$). When benign distributions are near-homogeneous under this file-level partition, the shared threshold is already a good fit and per-client calibration introduces variance without benefit—no personalization improvement was observed in this setting.

Interpretation caveat. Regime B uses file-defined CICIoT2023 pseudo-clients (dataset-file partitions, not physical IoT devices); the null result should not be generalized to physical-device partitions of CICIoT2023. Under this specific file-level partition, benign reconstruction-error distributions are near-homogeneous (mean JS divergence 0.004, max 0.038), so this result is a homogeneity-screen outcome rather than a general CICIoT2023 conclusion; alternative partitions (e.g., by attacker identity, attack type, or physical device) could yield different heterogeneity and should be evaluated before generalizing. All runs were budget-capped at 150 rounds (final-window relative loss change: 0.021–0.035, above the 0.005 convergence threshold). Comparisons remain controlled because all policies share the same frozen AE per seed and mean Macro-F1 remains stable at approximately 0.959 across all threshold strategies, indicating the score artifacts are sufficiently stable for threshold-scope comparison despite the budget cap.

E. Regime C: Non-IID Severity Sweep

Fig. 4 and Table VIII show that the personalization benefit is broadly consistent with larger gains under stronger heterogeneity, concentrated in the low-α band and absent under near-IID conditions. The $\alpha = 0.1$ and $\alpha = 0.3$ seed ranges overlap ($[0.535, 0.786]$ vs. $[0.420, 0.897]$), so strict monotonicity between these two levels should not be inferred

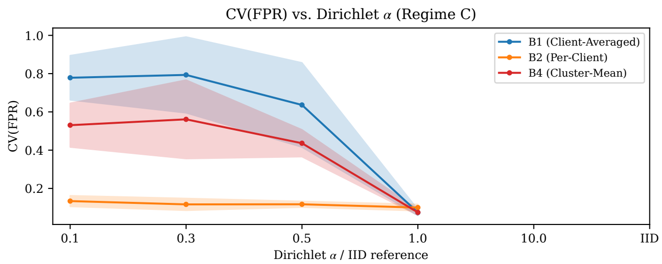

Fig. 4. Regime C: mean CV(FPR) by threshold strategy as a function of Dirichlet α. Whiskers show ± 1 std over 5 seeds. The B1–B2 gap is largest in the low-α band and disappears under IID or near-homogeneous conditions.

TABLE VIII
REGIME C (N-BAIOT DIRICHLET PARTITION, $K = 20$): MEAN CV(FPR) $\pm$ STD OVER 5 SEEDS.

α	B1 CV	B2 CV	B4 CV	Δ	Seed range
0.1	0.779 ± 0.104	0.134 ± 0.025	0.531 ± 0.103	0.645	[0.535, 0.786]
0.3	0.793 ± 0.178	0.117 ± 0.028	0.561 ± 0.184	0.677	[0.420, 0.897]
0.5	0.636 ± 0.197	0.117 ± 0.014	0.436 ± 0.063	0.519	[0.240, 0.777]
1.0	0.544 ± 0.116	0.096 ± 0.021	0.426 ± 0.066	0.448	[0.238, 0.565]
10.0	0.174 ± 0.053	0.111 ± 0.016	0.133 ± 0.012	0.063	[0.006, 0.122]
IID	0.074 ± 0.014	0.100 ± 0.015	0.075 ± 0.013	-0.026	[-0.032, -0.019]

Δ = B1 − B2 mean delta. Seed range = descriptive min–max over five seeds, not a confidence interval.

from this data; they form a high-heterogeneity band rather than a strictly ordered pair. B4 follows the same qualitative pattern but remains a partial approximation.

VII. Threats to Validity

FPR–TPR tradeoff. B2 reduces CV(FPR) but increases CV(TPR) (0.336 vs. 0.289) and decreases P10 Macro-F1 (0.300 vs. 0.344); worst-client balanced accuracy is not materially degraded. In this dataset, the P10 Macro-F1 degradation is concentrated in the low-separability Ennio Doorbell client (benign-vs-attack AUROC 0.931, lowest among all nine clients), so the result is best interpreted as a client-specific failure mode of percentile calibration rather than evidence of a universal FPR–TPR tradeoff. Per-client threshold personalization improves false-alarm dispersion fairness, not universal detection quality.

Dataset scope. Regime A uses 9 physical IoT devices with natural device-level heterogeneity (confirmatory); Regime C (K=20 synthetic clients) tests severity variation; Regime B provides a contrasting near-homogeneous applicability boundary. The results are a controlled measurement of threshold-scope effects under the tested protocol, not fleet-scale deployment validation; larger multi-vendor fleets are future work.

CICIoT2023 partition. The Regime B null result is an applicability boundary under our file-level CICIoT2023 pseudo-client partition, not evidence that threshold personalization is unnecessary for CICIoT2023 generally. A physical-device, attack-source, or deployment-identity partition could yield different heterogeneity and different threshold-personalization outcomes.

Calibration data sufficiency. In all reported conditions, calibration-pending count is zero. In deployments with fewer than $n_{\min} = 100$ benign calibration samples per client, B2 and B4 fall back to τ_{global}, reducing their advantage.

Threshold quantile sensitivity. q is fixed at 0.95. A threshold-only sensitivity check over $q \in \{0.90, 0.95, 0.99\}$ preserves the qualitative B2 < B4 < B1 ordering; absolute CV(FPR) values shift with q. The B1-pooled and B1-weighted shared-threshold sensitivity checks are consistent with the B1-to-B2 gap not being driven solely by the arithmetic-mean construction (Table VI).

Attack scope and adversarial robustness. The evaluation uses Mirai and BASHLITE variants on N-BaIoT. No poisoning, backdoor, or evasion evaluation is performed.

Privacy. No formal differential-privacy or cryptographic guarantees are provided. B4's 4-scalar fingerprint $[\mu_e, \sigma_e, \text{skew}_e, p_{95}(e)]$ constitutes distributional metadata leakage (not raw traffic), which could reveal coarse device-behavior class or tail characteristics; B4 is not a privacy mechanism. Mitigations (secure aggregation, differential privacy) are future work.

Convergence and budget-capped runs. Budget-capped runs limit broad convergence interpretation, but within each seed the same frozen AE is used across B1–B4, preserving threshold-calibration-only attribution.

Architecture and aggregation. The study uses a fixed AE $[d, 80, 40, 20]$ with vanilla FedAvg (E=1, full client participation every round). With E=1, FedAvg approaches synchronous distributed training with frequent aggregation, intentionally limiting local weight drift. Our results therefore characterize threshold-scope personalization under device-level score heterogeneity with a frequently aggregated encoder; they should not be read as evidence that the same improvement magnitude necessarily holds under drift-heavy FedAvg with larger local epochs. Larger E or different architectures may alter reconstruction-error geometry and the personalization benefit magnitude; this is a sensitivity direction for future work.

Concept drift. This study uses fixed train/calibration/test splits and does not model online recalibration; DATP assumes benign reconstruction-error distributions remain sufficiently stable between calibration and deployment.

Seeds and statistical power. Five seeds provide limited statistical power. Seed-level descriptive summaries (all deltas positive, range reported) are appropriate for this sample size but the study is underpowered for small effect sizes. All claims are scoped to the tested datasets, partitions, and protocol.

VIII. Conclusion

This study holds the FedAvg AE (E=1, full participation), training protocol, and seeds fixed across B1–B4; only the post-training threshold calibration scope varies. On heterogeneous N-BaIoT devices, per-client calibration (B2) provides the strongest FPR-dispersion reduction ($\Delta = 0.718$, all five seed deltas positive, range [0.588, 0.772]); cluster-mean calibration (B4) recovers a meaningful but partial share ($\approx$52% of B2's improvement) without requiring a device taxonomy, at the cost of transmitting distributional metadata that is not private. Family-mean calibration (B3) performs poorly: coarse device-type labels do not align with reconstruction-error calibration structure.

When benign error distributions are near-homogeneous (CICIoT2023 under our file-level pseudo-client partition, Dirichlet near-IID), no dispersion reduction is observed—this is an applicability boundary rather than evidence that personalization is unnecessary for these datasets generally. The main tradeoff under heterogeneous conditions is that B2 reduces FPR dispersion and improves mean Macro-F1, but the lower-tail P10 Macro-F1 degradation is concentrated in the low-separability Ennio Doorbell client—a client-specific failure of percentile calibration rather than evidence of a universal FPR–TPR tradeoff. Per-client calibration is an FPR-equity intervention, not a uniform detection-quality improvement.

Practitioner guidance. First, measure benign calibration-error heterogeneity across clients, for example using pairwise distribution divergence or dispersion of local calibration thresholds. Second, choose the threshold policy according to that heterogeneity and deployment constraints: use B1 when clients are near-homogeneous or per-client state is undesirable; use B2 when local calibration is available and false-alarm equity is the priority; use B4 when taxonomy-free grouping is acceptable, while accounting for the metadata leakage introduced by shared distributional fingerprints.

Limitations. Regime A uses only 9 physical devices; Regime B uses file-level pseudo-clients, not physical IoT devices; B4 fingerprints leak distributional metadata; no adversarial robustness, formal privacy, or real hardware evaluation is provided; five seeds limit statistical power (seed-level descriptive summaries are reported rather than confidence intervals). Because $E{=}1$ and all clients participate in every round, local weight drift is intentionally limited; evaluating whether the same threshold-personalization gains persist under larger local-epoch settings (which increase client drift and alter reconstruction-error geometry) is left as a sensitivity direction. A physical-device or attack-source CICIoT2023 partition would be required to test DATP under stronger CICIoT2023 heterogeneity; such a partition is outside the present controlled study. Threshold personalization is complementary to—not a replacement for—model-level personalization.

Future work: larger multi-vendor physical fleets, alternative CICIoT2023 partitions (physical-device identity, attack-source, attack-type), sensitivity to larger local-epoch settings ($E > 1$), privacy-preserving grouping (secure aggregation or differential privacy for fingerprint transmission), model-level personalization comparison, adversarial robustness under poisoning and evasion, and deployment validation under concept drift.

Availability. A replication package containing source code, configuration files, per-seed results, and table/figure generation scripts is archived on Zenodo (DOI: 10.5281/zenodo.20315654). Detailed reproduction commands are provided in the repository documentation.

References

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*. PMLR, 2017, pp. 1273–1282.

[2] Y. Meidan, M. Bohadana, Y. Mathov, Y. Mirsky, A. Shabtai, D. Breitenbacher, and Y. Elovici, "N-BaIoT: Network-based detection of IoT botnet attacks using deep autoencoders," *IEEE Pervasive Computing*, vol. 17, no. 3, pp. 12–22, 2018.

[3] E. C. Neto, S. Dadkhah, R. Ferreira, A. Bhaskaran, P. Ray, S. Dhouib, and A. A. Ghorbani, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environments," *Sensors*, vol. 23, no. 13, p. 5941, 2023.

[4] D. J. Beutel *et al.*, "Flower: A friendly federated learning research framework," *arXiv preprint arXiv:2007.14390*, 2020.

[5] V. Rey, P. M. Sánchez Sánchez, A. Huertas Celdrán, and G. Bovet, "Federated learning for malware detection in IoT devices," *Computer Networks*, vol. 204, p. 108693, 2022.

[6] A. Khraisat, A. Alazab, M. Alazab, A. Obeidat, S. Singh, and T. Jan, "Federated learning for intrusion detection in IoT environments: A privacy-preserving strategy," *Discover Internet of Things*, vol. 5, p. 72, 2025, doi: 10.1007/s43926-025-00169-7.

[7] B. Olanrewaju-George and B. Pranggono, "Federated learning-based intrusion detection system for the internet of things using unsupervised and supervised deep learning models," *Cyber Security and Applications*, vol. 3, p. 100068, 2025, doi: 10.1016/j.csa.2024.100068.

[8] V. T. Nguyen and R. Beuran, "FedMSE: Semi-supervised federated learning approach for IoT network intrusion detection," *Computers & Security*, vol. 151, p. 104337, 2025, doi: 10.1016/j.cose.2025.104337.

[9] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-IID data," *arXiv preprint arXiv:1806.00582*, 2018.

[10] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated learning: Challenges, methods, and future directions," *IEEE Signal Processing Magazine*, vol. 37, no. 3, pp. 50–60, 2020.

[11] A. Fallah, A. Mokhtari, and A. Ozdaglar, "Personalized federated learning with theoretical guarantees: A model-agnostic meta-learning approach," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 3557–3568.

[12] M. G. Arivazhagan, V. Aggarwal, A. K. Singh, and S. Choudhary, "Federated learning with personalization layers," *arXiv preprint arXiv:1912.00818*, 2019.

[13] P. P. Liang, T. Liu, L. Ziyin, N. B. Allen, R. P. Auerbach, D. Brent, R. Salakhutdinov, and L.-P. Morency, "Think locally, act globally: Federated learning with local and global representations," in *Workshop on Federated Learning for User Privacy and Data Confidentiality (NeurIPS)*, 2020.

[14] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: An ensemble of autoencoders for online network intrusion detection," in *Network and Distributed System Security Symposium (NDSS)*, 2018.

[15] T. D. Nguyen, S. Marchal, M. Miettinen, H. Fereidooni, N. Asokan, and A.-R. Sadeghi, "DIoT: Self-learning system for detecting compromised IoT devices," *IEEE Transactions on Information Forensics and Security*, vol. 14, no. 12, pp. 3314–3327, 2019.